# Satellite Orbit Design Contest Qualification Round Rules

## 1 Contest Objectives

This contest aims to develop participants' practical skills in **orbit design**, **space mission analysis**, and **orbital maneuver planning**. Each team must design a maneuver strategy for the "Human Spacecraft" (SC_Human) to intercept the "Alien Spacecraft" (SC_Alien) within a specified time limit, while also considering fuel consumption and mission efficiency.

## 2 Mission Setup

- **SC_Alien:** Subject only to gravitational forces; no active maneuvers are performed.
- **SC_Human:** May perform multiple instantaneous impulsive maneuvers ($\Delta\mathbf{v}_k$).
- **Maneuver constraints:**
  1. Each impulse must satisfy the velocity-increment limit:
     $$\|\Delta\mathbf{v}_k\| \leq \Delta V_{\text{lim}}.$$
     In this contest, the magnitude of each impulse must not exceed 250 m/s, i.e., $\Delta V_{\text{lim}} = 250$ m/s. Consecutive maneuvers must be separated by at least 100 seconds.
  2. The first maneuver may be performed at the beginning of the mission, if necessary.
  3. The total velocity increment $\Delta V_{\text{team}}$ must not exceed 3 km/s.
- **Trajectory constraint:** Before intercepting SC_Alien, the trajectory of SC_Human must not pass through the Earth.

## 3 Successful Interception Criteria

The relative distance is defined as

$$\Delta r(t) = \|\mathbf{r}_B(t) - \mathbf{r}_A(t)\|.$$

An interception is considered successful if there exists a time $t^*$ within $0 \leq t \leq T_{\text{max}}$ such that $\Delta r(t^*) \leq 5$ **km**. Here, $T_{\text{max}}$ denotes the mission end time, when the alien spacecraft is deemed to have escaped. Specifically, it is the first time after SC_Alien passes periapsis that its geocentric distance increases to the mission boundary. The value of $T_{\text{max}}$ will be announced at the start of the contest.

- **Interception time** ($t_{\text{int}}$): The **earliest** time at which the criterion is satisfied. If the qualifying times are $t^* = \{t_1^*, t_2^*, \ldots\}$, then $t_{\text{int}} = t_1^*$.
- **Mission completion time** ($T_{\text{team}}$): Either the interception time $t_{\text{int}}$ or the mission end time $T_{\text{max}}$.

Thus,

$$T_{\text{team}} = \begin{cases} t_{\text{int}}, & \text{successful interception}, \\ T_{\text{max}}, & \text{unsuccessful interception}, \end{cases} \tag{1}$$

$$d_{\text{min,team}} = \min_{0 \le t \le T_{\text{max}}} \Delta r(t), \tag{2}$$

$$\Delta r_{\text{min}} = \max\left(d_{\text{min,team}}, 5\right). \tag{3}$$

**Note:** This qualification round considers relative position only; **no restriction is imposed on the relative velocity** at closest approach.

# 4 Scoring Method

The total velocity increment $\Delta V_{\text{team}}$ is defined as

$$\Delta V_{\text{team}} = \sum_{k=1}^{N} \|\Delta \mathbf{V}_k\| \le 3 \text{ km/s}, \tag{4}$$

where $N$ is the total number of maneuvers.

The score accounts for interception performance, fuel efficiency, and time efficiency:

$$\begin{aligned} \text{Score} = 50 \exp\left(-\frac{\Delta r_{\text{min}} - 5}{100}\right) + \frac{25}{1 + e^{k_t(T_{\text{team}} - C_t)}} \\ + \frac{25}{1 + e^{k_v(\Delta V_{\text{team}} - C_v)}} - \sum_n P_n. \end{aligned} \tag{5}$$

The parameters $(k_t, C_t, k_v, C_v)$ depend on the orbital configuration and will be announced at the time of the contest.

**Important:** The following provisions apply to violations of the contest rules:

1. (a) If the velocity increment of any maneuver satisfies $\Delta V_k > \Delta V_{\text{lim}} + \epsilon_v$, the mission is deemed unsuccessful and the team will be excluded from the rankings. Here, $\epsilon_v = 10^{-5}$ m/s is the allowable numerical tolerance.

   (b) If the trajectory passes through the Earth before intercepting SC_Alien, the mission is deemed unsuccessful and the team will be excluded from the rankings.

   (c) If $\Delta V_{\text{team}} > 3$ km/s, the mission is deemed unsuccessful and the team will be excluded from the rankings.

   (d) For each pair of consecutive maneuvers separated by less than 100 seconds ($\Delta t < 100$ s), 10 points will be deducted ($P_n = 10$ per violation), with the final score floored at zero.

# 5 Tie-Breaking Criteria

If two or more teams receive the same score, rankings will be determined in the following order of priority:

| Priority | Criterion | Tie-breaking rule |
|---|---|---|
| 1 | Minimum relative distance $d_{\text{min,team}}$ | The smaller distance takes precedence. |
| 2 | Total velocity increment $\Delta V_{\text{team}}$ | The lower total velocity increment takes precedence. |
| 3 | Mission completion time $T_{\text{team}}$ | The shorter completion time takes precedence. |
| 4 | Design methodology | Tied teams shall each deliver a five-minute presentation explaining their orbit design approach. |

# 6 Qualification Results and Advancement

1. Only the top four teams in the qualification round will be announced. These teams will advance to the subsequent Gold Medal Match or Bronze Medal Match. Teams outside the top four will not be ranked individually and will all receive an Honorable Mention.
2. The top two teams will advance to the Gold Medal Match; the third- and fourth-place teams will advance to the Bronze Medal Match.
3. The rules for the Gold Medal Match and Bronze Medal Match will be announced separately.

# 7 Additional Provisions

- All results are subject to the verification program used by the organizer.
- External perturbations will be included in the qualification round, using the following models:
  - Gravity field: EGM96, degree 20 and order 20.
  - Tides: Tide File, model SolidAndPole.
  - Atmospheric model: NRLMSISE00; drag model: SPADFile.
  - Third-body gravity: Sun and Moon (Luna); solar radiation pressure using the SPADFile model; relativistic effects included.
- The organizer reserves the right to disqualify results that cannot be reproduced.
- Any amendments to these rules will be announced on the Taiwan Space Agency contest website. The latest published version shall prevail.
- All team members must be present during the final contest events (including the qualification round and the Gold and Bronze Medal Matches). Attendance by faculty advisors is not mandatory. However, if two teams remain tied after all tie-breaking criteria have been applied, the team whose faculty advisor is present will receive the higher ranking.

- The organizer reserves the right of final interpretation of these rules.